

JI-HOON KIM

Center for Theoretical Physics, Department of Physics & Astronomy
Institute for Data Innovation in Science
Seoul National University, Seoul 08826, Republic of Korea

www.jihoonkim.org
mornkr@snu.ac.kr

EDUCATION:

09/2005 - 06/2011 Ph. D., Department of Physics, Stanford University
03/1998 - 02/2002 B. Sc., School of Physics, *summa cum laude*, Seoul National University (SNU)

PROFESSIONAL APPOINTMENTS:

03/2024 - Team Leader, Center for Data-driven Research for Materials and Universe,
Institute for Data Innovation in Science, Seoul National University
03/2022 - **Associate Professor**, Seoul National University
03/2018 - 02/2022 Assistant Professor, Seoul National University
07/2016 - 11/2017 **Research Associate**, Stanford University / KIPAC
07/2015 - 06/2017 **Einstein Fellow**, Stanford University / SLAC National Accelerator Laboratory
07/2014 - 06/2015 Einstein Fellow, California Institute of Technology
01/2014 - 06/2014 **Moore Fellow**, California Institute of Technology
11/2013 - 12/2013 Visiting Scholar, Stanford University / KIPAC
09/2011 - 10/2013 **IMPS Postdoctoral Fellow**, University of California at Santa Cruz
09/2005 - 08/2011 Research Assistant, Stanford University

AWARDS AND FELLOWSHIPS (SELECTED):

03/2022 College of Natural Sciences Research Award, Seoul National University
03/2021 College of Natural Sciences Education Award, Seoul National University
01/2020 - 12/2022 POSCO Science Fellowship, POSCO TJ Park Foundation
09/2019 College of Natural Sciences Best Lecture Award, Seoul National University
12/2018 - 11/2023 Samsung Science & Technology Foundation Investigator
07/2014 - 06/2017 Einstein Postdoctoral Fellowship, NASA
01/2014 - 06/2014 Moore Postdoctoral Fellowship, California Institute of Technology
09/2005 - 03/2009 William R. and Sara Hart Kimball Graduate Fellowship, Stanford University
02/2002 Honor at Graduation, Seoul National University Alumni Association
03/2000 - 02/2002 Korea Foundation for Advanced Studies Fellowship

RESEARCH EXPERIENCES:

2022 - **Associate Professor**, Dept. of Physics and Astronomy, Seoul National University

- **AGORA Project Coordinator**, *Leading an inter-institutional collaboration for high-resolution simulations comparison since 2012* (160+ participants from 60+ institutions), having led or co-led 9+ papers, helped organize 16 workshops with 30+ attendees, and 150+ online conferences
- **PI, Samsung Science & Technology Foundation** “Towards the Multi-scale Understanding

of the Growth of Supermassive Black Holes” (12/2018-11/2023)

- **PI, KISTI KSC** “*Inferring Cosmological Parameters on Large-scale Structures via Domain-Generalized Neural Networks and Lightcone Simulations*”
(21,760,000 SUs on Nurion@KSC, Neuron@KSC, 20,000.0 GB on Nurion@KSC, 09/2024-08/2025)
- **PI, KISTI KSC** “*Multi-scale Numerical Experiment on Growth of SMBHs In Nuclear SCs*”
(4,352,000 SUs on Nurion@KSC, Neuron@KSC, 20,000.0 GB on Nurion@KSC, 01/2023-12/2023)
- **PI, KISTI KSC** “*Multi-scale Numerical Experiment on Growth of SMBHs Enhanced by TDEs*”
(6,528,000 SUs on Nurion@KSC, 20,000.0 GB on Nurion@KSC, 01/2022-12/2022)
- *Studying the co-evolution of massive black holes, nuclear star clusters, and their host galaxies*

2018 - 2022 **Assistant Professor**, Dept. of Physics and Astronomy, Seoul National University

- **PI, Creative-Pioneering Researchers Program at SNU** “*Towards the Multi-scale Understanding of the Growth of Supermassive Black Holes*” (09/2018-08/2019)
- **PI, KISTI KSC** “*Multi-scale Numerical Experiment on Growth of SMBHs and Their Environment*”
(7,834,000 SUs on Nurion@KSC, 20,000.0 GB on Nurion@KSC, 01/2021-12/2021)
- **PI, KISTI KSC** “*Multi-scale Numerical Experiment of Co-evolution of SMBHs and Their Hosts*”
(6,528,000 SUs on Nurion@KSC, 15,000.0 GB on Nurion@KSC, 01/2020-12/2020)
- **PI, KISTI KSC** “*Towards the Multi-scale Numerical Understanding of SMBHs at Galaxy Centers*”
(4,787,000 SUs on Nurion@KSC, 15,000.0 GB on Nurion@KSC, 12/2018-11/2019)
- **PI, KISTI KSC** “*Towards the Understanding of Growth & Evolution of SMBHs at Galaxy Centers*”
(400,000 SUs on Tachyon-II@KSC, 2,000.0 GB on Tachyon-II@KSC, 07/2018-10/2018)
- **Co-I, NASA HEC** “*FIRE: DM and Galaxy Formation with Unprecedented Physics and Resolution*”
(3,166,000 SUs on Pleiades@NAS, 100,000.0 GB on Pleiades@NAS, 11/2016-09/2019)
- *Simulating high-redshift quasar hosts with massive black holes and star-forming molecular clouds*

2015 - 2017 **Research Associate / Postdoctoral Researcher**, Stanford University / SLAC (Mentor: T. Abel)

- **PI, NSF XSEDE** “*Resolving the Impact of Supermassive Black Hole & Stellar Physics on Galaxies*”
(3,234,000 SUs on Stampede@TACC, 50,000.0 GB on Ranch@TACC, 10/2015-09/2016)
- **Co-I, NSF XSEDE** “*Simulating the Local Group*”
(3,649,000 SUs on Stampede@TACC, 50,000.0 GB on Ranch@TACC, 10/2016-09/2017)
- *Modeling the accretion and feedback of massive black hole seeds in the high-redshift universe*

2014 - 2015 **Postdoctoral Researcher**, California Institute of Technology (Mentor: P. Hopkins)

- **PI, NSF XSEDE** “*Resolving the Impact of Supermassive Black Holes on Galaxies*”
(1,200,000 SUs on Stampede@TACC, 20,000.0 GB on Ranch@TACC, 10/2014-09/2015)
- *Simulating and analyzing the formation of star clusters in high-redshift proto-galaxies*

2013 Visiting Scholar, KIPAC / Stanford University

2011 - 2013 **Postdoctoral Researcher**, UC Santa Cruz (Mentors: M. Krumholz & J. Primack)

- **Co-I, NSF XSEDE** “*Star Formation in Galaxies: From Recipes to Real Physics*”
(on Stampede/Ranger@TACC, Pleiades@NASA, Kraken@NICS)
- *Modeling radiative feedback of star-forming molecular clouds in galaxy formation*

- 2006 - 2011 **Research Assistant**, KIPAC/SLAC/Stanford University (Advisor: T. Abel)
- *Modeling the accretion and feedback of massive black holes, formation and feedback of stars*
- 2006 Research Assistant, KIPAC/Stanford University (Advisor: S. Church)
- *Building data pipelines for the QUAD Collaboration*
- 2005 Research Assistant, KIPAC/Stanford University (Advisor: E. Bloom)
- *Studying the cross-correlation between GRBs and SNe Ic events for the Fermi/GLAST Collaboration*

PEER-REVIEWED PUBLICATIONS [* = corresponding author] :

- **42)** Kim, Y., Hwang, H. S., Hong, S. E., Jeong, D., Yang, M., & **Kim, J. -H.**, “Three-dimensional Dark Matter Mapping in the Solar Neighborhood with Gaia Data Release 3 and Deep Learning”, *MNRAS submitted* (2026)
- **41)** Barrow, K., Nguyen, T. H., Roca-Fabrega, S., **Kim, J. -H.**, & 27 other co-authors for the AGORA Collaboration, “The AGORA High-resolution Galaxy Simulations Comparison Project. XI: Solving the Non-Spherical Morphology and Evolution of Dark Matter Halos with Haskap Pie”, *astro-ph:2605.24097, ApJ submitted* (2026)
- **40)** Kim, D., Jo, Y., Hwang, H. S., **Kim, J. -H.**, Kim, J., & 24 other co-authors, “Constructing a Mock Galaxy Catalog for the All-Sky Spectroscopic Survey (A-SPEC) Using Machine-assisted Semi-Simulation Model”, *astro-ph:2607.06933, ApJS in press* (2026)
- **39)** Chung, E., Jo, Y., **Kim, J. -H.***, Jung, M., & Kwon, O. -K., “From Dense Gas Clouds to Supermassive Black Hole Seeds: Self-consistent Simulations of High-Redshift Star Cluster Evolution”, *astro-ph:2606.10000, ApJ 1004* (2026) 250
- **38)** Jo, Y., Jung, M., Bryan, G. L., Kim, S., **Kim, J. -H.**, & Lee, A., “Evolution of Nuclear Star Cluster in Dwarf Galaxy through Mergers and In-Situ Star Formation”, *astro-ph:2603.18242, ApJ 1005* (2026) 103
- **37)** Kim, H., **Kim, J. -H.***, Jung, M., Roca-Fabrega, S., & 21 other co-authors for the AGORA Collaboration, “The AGORA High-resolution Galaxy Simulations Comparison Project. X: Formation and Evolution of Galaxies at the High-redshift Frontier”, *astro-ph:2511.04435, ApJ 1000* (2026) 276
- **36)** Nguyen, T. H., Barrow, K., Jung, M., Rodriguez-Cardoso, R., Roca-Fabrega, S., **Kim, J. -H.**, & 19 other co-authors for the AGORA Collaboration, “The AGORA High-resolution Galaxy Simulations Comparison Project. IX - Part 2: Effects of a Major Galaxy Merger on the Stellar and Gas Morphology of a Milky Way-mass Galaxy Progenitor”, *astro-ph:2607.21710, MNRAS submitted* (2026)
- **35)** Nguyen, T. H., Barrow, K., Jung, M., Rodriguez-Cardoso, R., Roca-Fabrega, S., **Kim, J. -H.**, & 19 other co-authors for the AGORA Collaboration, “The AGORA High-resolution Galaxy Simulations Comparison Project. IX - Part 1: Effects of a Major Galaxy Merger on Star Formation of a Milky Way-mass Galaxy Progenitor”, *astro-ph:2607.21709, MNRAS submitted* (2026)
- **34)** Jung, M., **Kim, J. -H.***, Nguyen, T. H., Rodriguez-Cardoso, R., Roca-Fabrega, S., & 22 other co-authors for the AGORA Collaboration, “The AGORA High-resolution Galaxy Simulations Comparison Project. VIII: Disk Formation and Evolution of Simulated Milky Way Mass Galaxy Progenitors at $1 < z < 5$ ”, *astro-ph:2505.05720, ApJ 994* (2025) 245
- **33)** Rodriguez-Cardoso, R., Roca-Fabrega, S., Jung, M., Nguyen, T. H., **Kim, J. -H.**, & 17 other co-authors for the AGORA Collaboration, “The AGORA High-resolution Galaxy Simulations Comparison Project. VII: Satellite Quenching in Zoom-in Simulation of A Milky Way-mass Halo”, *astro-ph:2505.05844, A&A 698 A303* (2025)
- **32)** Lee, S., Lee, H. M., **Kim, J. -H.***, Spurzem, R., Hong, J., & Chung, E., “Formation and Evolution of Compact Binaries Containing Intermediate Mass Black Holes in Dense Star Clusters”, *astro-ph:2503.22109, ApJ 988* (2025) 15

- 31) Lee, J. -Y., **Kim, J. -H.***, Jung, M., & 6 other co-authors, “Inferring Cosmological Parameters on SDSS via Domain-Generalized Neural Networks and Lightcone Simulations”, *astro-ph:2409.02256, ApJ* 975 (2024) 38
- 30) Jo, Y., Kim, S., **Kim, J. -H.**, & Bryan, G. L., “Evolution of Star Clusters Within Galaxies Using Self-consistent Hybrid Hydro/ N -body Simulation”, *astro-ph:2408.03128, ApJ* 974 (2024) 193
- 29) Lee, J., Shin, E. -J., **Kim, J. -H.**, Shapiro, P. R., & Chung, E., “Multiple Beads-on-a-string: Dark Matter-Deficient Galaxy Formation In A Mini-bullet Satellite-satellite Galaxy Collision”, *astro-ph:2312.11350, ApJ* 966 (2024) 72
- 28) Jung, M., **Kim, J. -H.***, Oh, B. K., Hong, S. E., Lee, J., & Kim, J., “Merger-tree-based Galaxy Matching: A Comparative Study Across Different Resolutions”, *astro-ph:2312.02466, ApJ* 965 (2024) 156
- 27) Strawn, C., Roca-Fabrega, S., Primack, J. R., **Kim, J. -H.***, & 24 other co-authors for the AGORA Collaboration, “The AGORA High-resolution Galaxy Simulations Comparison Project. VI: Similarities and Differences in the Circumgalactic Medium”, *astro-ph:2402.05246, ApJ* 962 (2024) 29
- 26) Jung, M., Roca-Fabrega, S., **Kim, J. -H.***, & 18 other co-authors for the AGORA Collaboration, “The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo”, *astro-ph:2402.05392, ApJ* 964 (2024) 123
- 25) Roca-Fabrega, S., **Kim, J. -H.***, Primack, J. R., Jung, M., & 23 other co-authors for the AGORA Collaboration, “The AGORA High-resolution Galaxy Simulations Comparison Project. IV: Halo and Galaxy Mass Assembly in a Cosmological Zoom-in Simulation at $z \leq 2$ ”, *astro-ph:2402.06202, ApJ* 968 (2024) 125
- 24) Jo, Y. et al. including **Kim, J. -H.**, “Calibrating Cosmological Simulations with Implicit Likelihood Inference Using Galaxy Growth Observables”, *astro-ph:2211.16461, ApJ* 944 (2023) 67
- 23) Lee, S., **Kim, J. -H.***, & Oh, B. K., “Growth of A Massive Black Hole In A Dense Star Cluster Via Tidal Disruption Accretion”, *astro-ph:2211.02376, ApJ* 943 (2023) 77
- 22) Shin, E. -J., Tacchella, S., **Kim, J. -H.**, Iyer, K. G., & Semenov, V. A., “Star Formation Variability As A Probe For the Baryon Cycle Within Galaxies”, *astro-ph:2211.01922, ApJ* 947 (2023) 61
- 21) Oh, B. K., An, H., Shin, E. -J., **Kim, J. -H.***, & Hong, S. E., “Machine-guided Exploration and Calibration of Astrophysical Simulations”, *astro-ph:2203.06914, MNRAS* 515 (2022) 693
- 20) Lee, J., Shin, E. -J., & **Kim, J. -H.***, “Dark Matter Deficient Galaxies And Their Member Star Clusters Form Simultaneously During High-velocity Galaxy Collisions in 1.25 pc Resolution Simulations”, *astro-ph:2108.01102, ApJ Letters* 917 (2021) L15
- 19) Roca-Fabrega, S., **Kim, J. -H.***, Hausammann, L., Nagamine, K., & 19 other co-authors for the AGORA Collaboration, “The AGORA High-resolution Galaxy Simulations Comparison Project. III: Cosmological Zoom-in Simulation of A Milky Way-mass Halo”, *astro-ph:2106.09738, ApJ* 917 (2021) 64
- 18) Shin, E. -J., **Kim, J. -H.***, & Oh, B. K., “How Metals Are Transported In And Out Of A Galactic Disk: Dependence On The Hydrodynamic Schemes In Numerical Simulations”, *astro-ph:2106.04640, ApJ* 917 (2021) 12
- 17) Shin, E. -J., Jung, M., Kwon, G., **Kim, J. -H.***, Lee, J., Jo, Y., & Oh, B. K., “Dark Matter Deficient Galaxies Produced Via High-velocity Galaxy Collisions in High-resolution Numerical Simulations”, *astro-ph:2007.09889, ApJ* 899 (2020) 25
- 16) Ma, X. et al. including **Kim, J. -H.**, “Self-consistent Proto-Globular Cluster Formation in Cosmological Simulations of High-redshift Galaxies”, *astro-ph:1906.11261, MNRAS* 493 (2020) 4315
- 15) **Kim, J. -H.***, Wise, J. H., Abel, T., Jo, Y., Primack, J. R., & Hopkins, P. F., “High-redshift Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Stellar Feedback and Quasar Growth”, *astro-ph:1910.12888, ApJ* 887 (2019) 120

- **14)** Jo, Y., & **Kim, J. -H.***, “Machine-assisted Semi-Simulation Model (MSSM): Estimating Galactic Baryonic Properties from Their Dark Matter Using A Machine Trained on Hydrodynamic Simulations”, *astro-ph:1908.09844*, *MNRAS* 489 (2019) 3565
- **13)** Bryan, G. L. et al. including **Kim, J. -H.** for the *ENZO* Collaboration, “*Enzo*: An Adaptive Mesh Refinement Code for Astrophysics (Version 2.6)”, *JOSS* 4(42) (2019) 1636
- **12)** **Kim, J. -H.***, Ma, X., Grudic, M. Y., Hopkins, P. F., Hayward, C. C., & 5 other co-authors for the *FIRE* Collaboration, “Formation of Globular Cluster Candidates in Merging Proto-galaxies at High Redshift: A View from the *FIRE* Cosmological Simulations”, *astro-ph:1704.02988*, *MNRAS* 474 (2018) 4232
- **11)** Hopkins, P. F. et al. including **Kim, J. -H.**, “The *FIRE*-2 Simulations: Physics Versus Numerics in Galaxy Formation”, *astro-ph:1702.06148*, *MNRAS* 480 (2018) 800
- **10)** Butsky, I., Zrake, J., **Kim, J. -H.**, Yang, H. -I., & Abel, T., “Ab Initio Simulations of A Supernova Driven Galactic Dynamo in An Isolated Galaxy”, *astro-ph:1610.08528*, *ApJ* 843 (2017) 113
- **9)** **Kim, J. -H.***, Agertz, O., Teyssier, R., Butler, M. J., Ceverino, D., & 38 other co-authors for the *AGORA* Collaboration, “The *AGORA* High-resolution Galaxy Simulations Comparison Project. II: Isolated Disk Test”, *astro-ph:1610.03066*, *ApJ* 833 (2016) 202
- **8)** Wetzel, A., Hopkins, P. F., **Kim, J. -H.**, Faucher-Giguere, C-A., Keres, D., & Quataert, E., “Reconciling Dwarf Galaxies with LCDM Cosmology: Simulating A Realistic Population of Satellites Around A Milky Way-Mass Galaxies”, *astro-ph:1602.05957*, *ApJ Letters* 827 (2016) L23
- **7)** **Kim, J. -H.***, Abel, T., Agertz, O., Bryan, G. L., Ceverino, D., & 41 other co-authors for the *AGORA* Collaboration, “The *AGORA* High-resolution Galaxy Simulations Comparison Project”, *astro-ph:1308.2669*, *ApJS* 210 (2014) 14
- **6)** Bryan, G. L. et al. including **Kim, J. -H.** for the *ENZO* Collaboration, “*Enzo*: An Adaptive Mesh Refinement Code for Astrophysics”, *astro-ph:1307.2265*, *ApJS* 211 (2014) 19
- **5)** **Kim, J. -H.***, & Lee, J., “How Does the Surface Density and Size of Disk Galaxies Measured in Hydrodynamic Simulations Correlate with the Halo Spin Parameter?”, *astro-ph:1210.8321*, *MNRAS* 432 (2013) 1701
- **4)** **Kim, J. -H.***, Krumholz, M. R., Wise, J. H., Turk, M. J., Goldbaum, N. J., & Abel, T., “Dwarf Galaxies with Ionizing Radiation Feedback. II: Spatially-resolved Star Formation Relation”, *astro-ph:1210.6988*, *ApJ* 779 (2013) 8
- **3)** **Kim, J. -H.***, Krumholz, M. R., Wise, J. H., Turk, M. J., Goldbaum, N. J., & Abel, T., “Dwarf Galaxies with Ionizing Radiation Feedback. I: Escape of Ionizing Photons”, *astro-ph:1210.3361*, *ApJ* 775 (2013) 109
- **2)** **Kim, J. -H.***, Wise, J. H., Alvarez, M. A., & Abel, T., “Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes. I: Feedback-regulated Star Formation and Black Hole Growth”, *astro-ph:1106.4007*, *ApJ* 738 (2011) 54
- **1)** **Kim, J. -H.***, Wise, J. H., & Abel, T., “Galaxy Mergers with Adaptive Mesh Refinement: Star Formation and Hot Gas Outflow”, *astro-ph:0902.3001*, *ApJ Letters* 694 (2009) L123

OTHER CONTRIBUTIONS:

- **7)** Roca-Fabrega, S., **Kim, J. -H.**, Primack, J. R., & 9 other co-authors for the *AGORA* Collaboration, “The *AGORA* High-resolution Galaxy Simulations Comparison Project: CosmoRun Public Data Release”, *astro-ph:2408.00432*
- **6)** Shin, E. -J., & **Kim, J. -H.***, “News and Views: Giant Collision Created Galaxies Devoid of Dark Matter”, *Nature* 605 (2022) 427
- **5)** Roca-Fabrega, S., **Kim, J. -H.**, Primack, J. R., & 11 other co-authors for the *AGORA* Collaboration, “The *AGORA* High-resolution Galaxy Simulations Comparison Project: Public Data Release”, *astro-ph:2001.04354*

- 4) Pineda, J. L. et al. including **Kim, J. -H.**, “Bridging the Gap: Observations and Theory of Star Formation Meet on Large and Small Scales”, *Keck Institute for Space Studies Report, Pasadena, CA, November 2014*
- 3) Kaehler, R., Abel, T., & **Kim, J. -H.**, “Visualization of a High-resolution Galaxy Formation Simulation”, *SuperComputing '11 Scientific Visualization Companion Proceedings*, pp. 133-134, Seattle, WA, November 2011
- 2) **Kim, J. -H.**, “Galaxy Formation and Mergers with Stars and Massive Black Holes”, *Doctoral Dissertation, Stanford University, Stanford, CA, May 2011*
- 1) **Kim, J. -H.**, Wise, J. H., & Abel, T., “Galaxy Evolution on Adaptive Mesh Refinement”, *First Stars III Conference, AIP Conference Proceedings*, Vol. 990 (2008) pp. 429-431, Santa Fe, NM, July 2007

PROFESSIONAL AND OUTREACH SERVICES:

- | | |
|-------------|--|
| 2025 | Organizer, 4th SNUARC Woojeon Astronomy Workshop, SNU (Aug. 21-22, 2025) |
| 2025 | Organizer , 13th Workshop for the AGORA Project at UC Santa Cruz (Aug. 8-9, 2025) |
| 2025 | Member, Scientific Organizing Committee for “International Workshop on Galaxy Formation + AGORA in Asia 2025”, Osaka, Japan (May 26-30, 2025) |
| 2024 | Organizer, 12th Workshop for the AGORA Project, online (Aug. 8, 2024) |
| 2024 | Organizer , 2nd CTP Bosan Workshop: AGORA in Asia + 5th Numerical Galaxy Formation Meeting, SNU (May 7-10, 2024) |
| 2024 - 2027 | Professional Committees Served: Academic Affairs/Korean Astronomical Society |
| 2023 | Member, Scientific Organizing Committee for “Galaxy Formation in Hangzhou: Observations and Physics of AGN Feedback”, Hangzhou, China (Oct. 10-13, 2023) |
| 2023 | Organizer, 11th Workshop for the AGORA Project at UC Santa Cruz (Aug. 11-12, 2023) |
| 2023 - | Organizer, Kyewon Physics Public Lecture, Center for Theoretical Physics, SNU |
| 2023 | Co-organizer, 4th Numerical Galaxy Formation Workshop at Konjiam (Jan. 11, 2023) |
| 2023 - 2026 | Member, Board of Directors , Korean Astronomical Society |
| 2022 | Organizer, 10th Workshop for the AGORA Project at UC Santa Cruz (Aug. 19-20, 2022) |
| 2022 | Mentor, SNU Science Education Institute for The Gifted (Physics) |
| 2022 | Organizer, 8th Korea Astronomy Machine Learning Meeting, online (Feb. 22, 2022) |
| 2022 | Speaker, SNU-Naver Lecture Series: One Thousand Keywords (Feb. 11, 2022) |
| 2022 | Organizer, 3rd Numerical Galaxy Formation Workshop, online (Jan. 25, 2022) |
| 2021 | Organizer, 9th Workshop for the AGORA Project, online (Aug. 12-13, 2021) |
| 2021 | Speaker, KAOS Foundation Science Concert (<i>Knowledge Awakening On Stage</i> ; Jul. 28, 2021) |
| 2021 | Co-organizer, 2nd Numerical Galaxy Formation Workshop, online (Jan. 28, 2021) |
| 2020 | Organizer, 1st Numerical Galaxy Formation Workshop, SNU (Jan. 16, 2020) |
| 2019 | Organizer, 8th Workshop for the AGORA Project at UC Santa Cruz (Aug. 9-10, 2019) |
| 2019 - 2022 | Member, National Organizing Committee for the International Astronomical Union General Assembly 2022, Busan, Korea (Aug. 2-11, 2022) |
| 2018 - | Member, Korea e-Science Forum, Korea Institute of Science and Technology Information |
| 2018 | Organizer, 7th Workshop for the AGORA Project at UC Santa Cruz (Aug. 10-11, 2018) |
| 2018 - | Department Committees Served: Graduate Admission and Qualification, Academic Affairs, Student Affairs and Scholarship, Department Planning, |

Faculty Search and Hiring

- 2018 - **Professional Memberships:** International Astronomical Union, Korean Physical Society, Korean Astronomical Society
- 2017 Organizer, 6th Workshop for the AGORA Project at UC Santa Cruz (Aug. 11-13, 2017)
- 2016 Expert Referee, DiRAC Resource Allocation Committee, the United Kingdom
- 2016 External Review Panel, NASA Postdoctoral Program Fellowship
- 2016 Organizer, 5th Workshop for the AGORA Project at UC Santa Cruz (Aug. 12-14, 2016)
- 2015 Organizer, 4th Workshop for the AGORA Project at UC Santa Cruz (Aug. 21-23, 2015)
- 2015 External Review Panel, Department of Energy ASCR Leadership Computing Challenge
- 2015 External Review Panel, NASA Earth and Space Science Fellowship
- 2014 Organizer, 3rd Workshop for the AGORA Project at UC Santa Cruz (Aug. 15-17, 2014)
- 2014 - **Referees Served:** Astrophysical Journal, Astrophysical Journal Letters, Monthly Notices of the Royal Astronomical Society, Astronomy and Astrophysics, Astronomy and Computing, Open Journal of Astrophysics, Journal of the Korean Astronomical Society
- 2013 Organizer, 2nd Workshop for the AGORA Project at UC Santa Cruz (Aug. 16-18, 2013)
- 2012 External Review Panel, NASA Earth and Space Science Fellowship
- 2012 Organizer, 1st Workshop for the AGORA Project at UC Santa Cruz (Aug. 17-19, 2012)
- 2012 - **Project Coordinator,** AGORA High-resolution Galaxy Simulations Comparison Project
- 2010 Provided a movie data of cosmological galaxy mergers and structure formation for the SLAC booth in the SuperComputing '10 conference (SC2010), New Orleans, LA
- 2010 Provided a movie data of cosmological galaxy mergers for the Hayden Planetarium at the American Museum of Natural History, NY in the program of "The Big Bang"
- 2007 - 2011 KIPAC/SLAC Visualization Laboratory Presenter, Laboratory Tour Guide
- 2007 - 2009 Volunteer, SLAC Kids' Day
- 2002 - 2004 Lieutenant, Company Commander & Battalion Staff Officer, the Republic of Korea Army

TEACHING EXPERIENCES:

- 2026 **Lecturer,** 1 semester, Seoul National University, *Mathematical Physics I*
- 2026 Lecturer, Korea Institute for Advanced Studies Astrophysics Summer School 2026, *Artificial Intelligence in Astronomy*
- 2026 **Lecturer,** 1 semester, Seoul National University, *Classical Mechanics I*
- 2026 Lecturer, Korea Physics Olympiad (KPhO) Winter School 2026, *Mechanics and Astrophysics*
- 2025 **Lecturer,** 1 semester, Seoul National University, *Mathematical Physics I*
- 2025 Lecturer, Korea Institute for Advanced Studies Astrophysics Summer School 2025, *Numerical Experiments in Astrophysics and Cosmology*
- 2025 Lecturer, Korea Physics Olympiad (KPhO) Winter School 2025, *Mechanics and Astrophysics*
- 2024 Lecturer, Korea Institute for Advanced Studies Astrophysics Summer School 2024, *Reproducibility in Astrophysical Simulations*
- 2024 **Lecturer,** 2 semesters, Seoul National University, *Physics I & II (for physics/astronomy majors)*

2023	Lecturer, Korea Institute for Advanced Studies Astrophysics Summer School 2023, <i>Back of the Envelope (Astro)Physics</i>
2023	Lecturer , 2 semesters, Seoul National University, <i>Physics I & II (for physics/astronomy majors)</i>
2022	Lecturer , 1 semester, Seoul National University, <i>Mathematical Physics I</i>
2022	Lecturer, Korea Institute for Advanced Studies Astrophysics Summer School 2022, <i>Supermassive Black Holes</i>
2022	Lecturer , 1 semester, Seoul National University, <i>Physics I</i>
2021	Lecturer , 1 semester, Seoul National University, <i>Mathematical Physics I</i>
2020	Lecturer , 2 semesters, Seoul National University, <i>Classical Mechanics I & II</i>
2019	Lecturer, Korea Institute for Advanced Studies - SNU Physics Winter Camp 2019, <i>Astrophysical Black Holes</i>
2019	Lecturer , 2 semesters, Seoul National University, <i>Classical Mechanics I & II</i>
2018	Lecturer, Korea Institute for Advanced Studies - SNU Physics Winter Camp 2018, <i>Astrophysical Dark Matter</i>
2018	Lecturer , 1 semester, Seoul National University, <i>Topics in Modern Astrophysics and Cosmology</i>
2018	Guest Lecturer, Seoul National University, <i>Classical Mechanics</i>
2016	Guest Lecturer, Stanford University, <i>Computational Cosmology and Astrophysics</i>
2014	Guest Lecturer, California Institute of Technology, <i>Cosmology and Galaxy Formation</i>
2009	Teaching Assistant, 1 quarter, Stanford University, <i>Black Holes</i>
2008	Course Grader, 1 quarter, Stanford University, <i>Introduction to Astrophysics</i>
2007	Teaching Assistant, 1 quarter, Stanford University, <i>Modern Physics Laboratory</i>
2007	Teaching Assistant, 1 quarter, Stanford University, <i>Physics</i>

MENTORING EXPERIENCES:

- Postdoctoral scholars : Dr. Keita Fukushima (10/2024 -),
Dr. Boon Kiat Oh (10/2019 - 09/2022, → postdoc @ University of Connecticut)
- Ph. D. students : Eunwoo Chung (09/2024 -), Hyeonyong Kim (09/2023 -),
Minyong Jung (09/2021 -), Seungjae Lee (09/2019 - 08/2025),
Eun-jin Shin (03/2018 - 08/2023, → postdoc @ University of Cambridge),
Yongseok Jo (03/2018 - 08/2022, → postdoc @ Columbia University & Flatiron Institute)
- M. Sci. students : Chan-mi Shin (09/2026 -), Yiyang Wang (09/2025 -), Hyeonu Seo (09/2025 -),
Jun Yong Park (09/2022 - 02/2026), Ki-won Kim (09/2018 - 08/2019)
- B. Sci. interns : Hyungjun Kim (06/2026 -), Minje Park (06/2026 - , from Sungkyunkwan U.),
Minjun Choi (12/2025 -), Seoyeon Park (12/2025 -),
Keonwoo Leo Kim (06/2025 - 08/2025, from McGill U.), Yiyang Wang (03/2025 - 07/2025),
Hyeonmin Eugene Lee (02/2025 - 08/2025, from UT Austin),
Daekyu Yoon (12/2024 -), Chaerin Jeong (08/2024 - 05/2025, from KyungHee U.),
Gyujin Kim (07/2024 - 04/2025), Geon Hee Kim (06/2024 - 02/2025),
Abdelrahman Helal (09/2023 - 12/2023, from Minerva U.),
Youngjun Cho (08/2023 - 11/2023, from UNIST),
Junhwa Jang (03/2023 - 06/2024, 03/2026 -), Hwan Hee Chung (03/2023 - 03/2024),
Donghwan Hyun (12/2022 - 02/2024, Thesis), Subo Hwang (12/2022 - 02/2024, Thesis),

Eunwoo Chung (12/2022 - 02/2024, Thesis),
Han Noh (07/2022 - 08/2022, from Andong U.),
Yong-uk Cho (06/2022 - 08/2022, from KyungHee U.),
Seoyoung Kim (12/2021 - 05/2024), Jun-Young Lee (12/2021 - 05/2025, Thesis),
Ho-sung Lee (09/2021 - 01/2022), Junhee Myong (06/2021 - 05/2023),
Ahram Lee (03/2021 - 05/2022), Hyeongmo Kim (01/2021 - 09/2021),
Minju Kum (12/2020 - 02/2021), Songyoun Park (10/2020 - 04/2022),
Hyeonyong Kim (09/2020 - 02/2023), Lael Shin (09/2020 - 03/2021, Thesis),
Eunhee Ko (09/2020 - 02/2021, Thesis), Su-un Lee (08/2020 - 10/2021, Thesis),
Hongjun An (08/2020 - 02/2021, Thesis), Hyerin Cho (01/2020 - 08/2020, from GIST),
Sangmin Bae (12/2019 - 07/2020), Eonho Chang (10/2019 - 12/2019, from UC Santa Cruz),
Joohyun Lee (09/2019 - 08/2021, Thesis), Seung-o Ha (07/2019 - 08/2019, from Postech),
Goojin Kwon (07/2019 - 09/2019, 06/2020 - 05/2021, 09/2021 - 11/2021, from U. Cambridge),
Minyong Jung (03/2019 - 08/2020), Sangmoon Lee (03/2019 - 08/2019, Thesis)

COMPUTING EXPERIENCES:

- Administrator of a Intel Xeon SP Gold + GPU cluster with 904 cores and max 1 TB memory / node (2018 -)
- Program experience: Enzo, Gadget, GIZMO, yt, MUSIC, VisIt, PartiView, HEALpix, CMBFast, etc.
- Language experience: C, C++, Fortran, IDL, Python, Java Applet, Visual Basic, Pascal, Matlab, etc.

RESEARCH INTERESTS:

- Coordinator, AGORA High-resolution Galaxy Simulations Comparison Project (AGORAsimulations.org)
- Galaxy Formation and Evolution using High-resolution Adaptive Mesh Refinement Simulations
- Radiative / Mechanical / Thermal Feedback from Massive Black Holes and Star-forming Molecular Clouds
- Growth of Supermassive Black Holes and Triggered Star Formation via Gas Inflow or Mergers

TALKS AND SEMINARS (SELECTED):

- **12th KIAS Workshop on Cosmology and Structure Formation**, *hosted by Korea Institute for Advanced Study, TBD, 11/2026*
- **Physics Colloquium**, *Korea Institute for Advanced Study, TBD, 10/2026*
- **XXXVII IUPAP Conference on Computational Physics (CCP2026)**, *Sejong University Convention Center, hosted by Korean Physical Society, TBD, 08/2026*
- **1st Galaxy Formation and AGORA Workshop in Europe**, *hosted by Lund University, "Cosmic Mystery of Super-early, Super-massive Black Holes: Perspectives from the Hydrodynamics+N-body Hybrid Simulations", 05/26/2026*
- **Institute of Astronomy (IoA) Colloquium**, *University of Cambridge, "Reproducibility in Astrophysical Simulations: The AGORA Initiative and a 14-Year Human Experiment (In Memory of Joel Primack)", 05/21/2026*
- **8th Chirality in the Universe Beyond the Electroweak Scale (CUBES) Workshop**, *hosted by Center for Underground Physics and Center for Theoretical Physics of the Universe, Institute for Basic Science, Korea, "Cosmic Mystery of Super-early, Super-massive Black Holes: Perspectives from the Hydrodynamics+N-body Hybrid Simulations", 04/25/2026*
- **BK21 Annual Forum**, *Seoul National University, "Cosmic Mystery of Super-early, Super-massive Black Holes: Perspectives in the Era of High-resolution Simulations", 12/11/2025*

- **4th SNUARC Astronomy Workshop**, *Seoul National University, hosted by SNU Astronomy Research Center, "Computational Cosmology Research"*, 08/21/2025
- **Santa Cruz Galaxy Formation Workshop+13th AGORA Workshop**, *UC Santa Cruz, "Past, Present, and Future of the AGORA Project (and a 13-Year Human Experiment)"*, 08/08/2025
- **IBS CTPU-CGA Summer Workshop 2025**, *hosted by Center for Theoretical Physics of the Universe (Cosmology, Gravity and Astroparticle Physics Group), Institute for Basic Science, Korea, "Cosmic Mystery of Supermassive Black Holes: Perspectives in the Era of High-resolution Simulations"*, 07/31/2025
- **International Workshop on Galaxy Formation+AGORA in Asia 2025**, *Nakanoshima Center, hosted by University of Osaka, "Simulating Dark Matter-Deficient Galaxy Formation in A Mini-bullet Satellite-satellite Galaxy Collision + The Status of the AGORA Project"*, 05/28/2025
- **Artificial Intelligence and High Energy Physics (AI+HEP) in East Asia**, *Center for Theoretical Physics of the Universe (Particle Theory and Cosmology Group), Institute for Basic Science, Korea, "How Can Machine Learning and Simulations Help Us To Study Our Universe?"*, 02/24/2025
- **11th KIAS Workshop on Cosmology and Structure Formation**, *hosted by Korea Institute for Advanced Study, "Reproducibility In Astrophysical Simulations and the AGORA Initiative (and a 12-Year Human Experiment)"*, 11/01/2024
- **Physics Colloquium**, *Hanyang University, "Solving the Mysteries of Supermassive Black Holes in the Era of High-resolution Numerical Simulations"*, 10/30/2024
- **Statistics Seminar**, *Seoul National University, "How Can Machine Learning Help Us When Studying Our Universe (And Astronomically Large Datasets)?"*, 10/11/2024
- **Symmetry and Structure of the Universe Workshop 2024**, *Jeonbuk National University, "New Physics Changes Massive Black Holes' Evolution: Perspectives in the Era of High-resolution Simulations"*, 07/29/2024
- **Physics Colloquium**, *Korea University, "New Physics Changes Massive Black Holes' Evolution: Perspectives in the Era of High-resolution Simulations"*, 04/02/2024
- **Inaugural Symposium of the Institute for Data Innovation in Science**, *Seoul National University, "How Can Machine Learning Help Us When Studying Our Universe?"*, 03/28/2024
- **Survey Science Group Workshop 2024**, *hosted by Korea Institute for Advanced Study, "AGORA High-resolution Galaxy Simulations Comparison Initiative: 12 Years After Its Conception (and a 12-Year Human Experiment)"*, 01/31/2024
- **Galaxy Formation in Hangzhou: Observations and Physics of AGN Feedback**, *hosted by Institute for Advanced Study in Physics, Zhejiang University and Shanghai Astronomical Observatory, Chinese Academy of Sciences, "How Including New Physics Changes Massive Black Holes' Evolution: Perspectives in the Era of High-resolution Simulations"*, 10/11/2023
- **Korea Supercomputing Conference 2023**, *hosted by Korea Institute of Science and Technology Information, "How Can Machine Learning Help Us When Studying and Simulating the Universe?"*, 08/22/2023
- **DARWIN+Dwarf Galaxy Workshop**, *Korea Astronomy and Space Science Institute, "How Can Machine Learning Help Us When Studying and Simulating the Universe?"*, 08/16/2023
- **Santa Cruz Galaxy Formation Workshop+11th AGORA Workshop**, *UC Santa Cruz, "Overview of the AGORA Project" (joint with Joel Primack and Santi Roca-Fabrega)*, 08/11/2023
- **XV International Conference on Gravitation, Astrophysics and Cosmology**, *hosted by Asia Pacific Center for Theoretical Physics, "How Including New Physics Changes Massive Black Holes' Evolution: Perspectives in the Era of High-resolution Simulations"*, 07/06/2023
- **Cosmology Workshop on the Crossroad of Astrophysics and Particle Physics**, *hosted by Korea Institute for Advanced Study, "Studying Dark Matter Using Realistic Simulations"*, 06/29/2023

- **Astronomy Colloquium**, *Chungnam National University*, "Taking on the Mysteries in the Universe in the Era of High-resolution Simulations", 06/07/2023
- **10th KIAS Workshop on Cosmology and Structure Formation**, *Korea Institute for Advanced Study*, "How Including New Physics Changes Galactic Evolution In Simulations: Perspectives in the Era of High-resolution Simulations", 10/28/2022
- **9th East Asian Numerical Astrophysics Meeting**, *Tenbusu Hall, Okinawa, hosted by Chiba University*, "How Feedback Affects Galactic Evolution: Perspectives in the Era of High-resolution Simulations", 09/26/2022
- **Santa Cruz Galaxy Formation Workshop+10th AGORA Workshop**, *UC Santa Cruz*, "The AGORA Project To Compare Zoom-in Cosmological Simulations By The Leading Hydro Codes of A MW-mass Galaxy" (joint with Joel Primack), 08/19/2022
- **International Astronomical Union Symposium 373: Resolving the Rise and Fall of Star Formation in Galaxies (at IAU General Assembly XXXI)**, *Busan Exhibition and Convention Center, co-hosted by Korean Astronomical Society and Korea Astronomy and Space Science Institute*, "How Feedback Affects Stellar and Galactic Evolution: Perspectives in the Era of High-resolution Simulations", 08/10/2022
- **Center for Theoretical Physics Colloquium**, *Seoul National University*, "Observational Astrophysics Primer To Prep For JWST Star Party", 07/20/2022
- **1st SNURC Astronomy Workshop**, *Seoul National University Siheung Convention Center, hosted by SNU Astronomy Research Center*, "Computational Cosmology Research", 06/30/2022
- **Saturday Science Open Lecture**, *Seoul National University*, "Studying Our Universe Like A Pro (III)", 04/09/2022
- **Beyond the Standard Model Workshop 2022**, *Chung-Ang University*, "Solving the Mysteries of Supermassive Black Holes in the Era of High-resolution Simulations", 02/10/2022
- **Physics Colloquium**, *Ajou University*, "Solving the Mysteries of Supermassive Black Holes in the Era of High-resolution Simulations", 12/08/2021
- **Annual Forum 2021**, *Samsung Science & Technology Foundation*, "Towards The Unabridged Understandings of The Growth of Supermassive Black Holes", 11/22/2021
- **Saturday Science Open Lecture**, *Seoul National University*, "Studying Our Universe Like A Pro (II)", 11/20/2021
- **Astronomy Colloquium**, *Shanghai Jiao Tong University*, "Solving the Mysteries of Galaxy and Supermassive Black Hole Formation in the Era of High-resolution Simulations", 10/20/2021
- **Astrophysics Seminar**, *Institute of Particle and Cosmos Physics, Universidad Complutense de Madrid*, "Solving the Mysteries of Galaxy and Supermassive Black Hole Formation in the Era of High-resolution Simulations and Machine Learning", 05/14/2021
- **Saturday Science Open Lecture**, *Seoul National University*, "Studying Our Universe Like A Pro", 05/01/2021
- **Physics Colloquium**, *Yonsei University*, "Solving the Mysteries of Supermassive Black Holes in the Era of High-resolution Simulations", 04/28/2021
- **7th Galaxy Evolution Workshop**, *Institute of Astronomy and Astrophysics, Academia Sinica, National Taiwan University*, "Solving the Mysteries of Galaxy and Supermassive Black Hole Formation in the Era of High-resolution Simulations and Machine Learning", 02/03/2021
- **College of Education Seminar**, *Chosun University*, "Exoplanets: From Mythology To Science", 12/23/2020
- **Physics Colloquium**, *Chung-Ang University*, "Solving the Mysteries of Supermassive Black Holes in the Era of High-resolution Simulations", 12/07/2020

- **Theoretical Astrophysics Colloquium**, *University of Osaka*, "Solving the Mysteries of Galaxy and Supermassive Black Hole Formation in the Era of High-resolution Simulations and Machine Learning", 10/21/2020
- **Numerical Astrophysics Workshop at Korea Supercomputing Conference 2020**, *hosted by Korea Institute of Science and Technology Information*, "Numerical Galaxy Formation in the Era of High-resolution Simulations and Machine Learning", 09/24/2020
- **Science and Policy Advanced Research Course (SPARC)**, *Seoul National University*, "Exoplanets: From Mythology To Science", 01/07/2020
- **Galaxy Formation and Evolution Across Cosmic Time**, *Institute of Astronomy and Astrophysics, Academia Sinica, National Taiwan University*, "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", 12/11/2019
- **Astronomy Colloquium**, *Yonsei University*, "Upcoming Era in Numerical Galaxy Formation: New Possibilities and Challenges", 12/03/2019
- **Center for Theoretical Physics Colloquium**, *Seoul National University*, "How To Make Your Own Universe On A Computer: Success and Challenges", 11/08/2019
- **XAIENCE: Crossing-over the AI and Science**, *Seoul National University*, "Machine Learning In Astrophysics & Cosmology: Estimating Galactic Baryonic Properties from Their Dark Matter", 11/07/2019
- **Special Physics Colloquium**, *Seoul National University*, "Cosmology and Exoplanets: How Scientists Turned Mythology Into Physics", 10/30/2019
- **Zurich Meets Seoul**, *co-hosted by University of Zurich and Seoul National University*, "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", 10/01/2019
- **Santa Cruz Galaxy Formation Workshop+8th AGORA Workshop**, *UC Santa Cruz*, "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", 08/08/2019, "AGORA Project Update: 7 Years After Conception" (*joint with Joel Primack & Santi Roca-Fabrega*), 08/09/2019
- **4th Korean-American Kavli Frontiers of Science Symposium**, *co-sponsored by the U.S. National Academy of Sciences (NAS) and the Korean Academy of Science and Technology (KAST)*, "Upcoming Era in Numerical Galaxy Formation: New Possibilities and Challenges", 06/19/2019
- **Astronomy Colloquium**, *Kyungpook National University*, "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", 06/04/2019
- **95th Korean Physical Society Meeting**, *hosted by KPS*, "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", 04/26/2019
- **Extremely Big Eyes on the Early Universe**, *Kavli Institute for the Physics and Mathematics of the Universe, University of Tokyo*, "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", 03/25/2019
- **Physics Colloquium**, *Seoul National University*, "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", 12/05/2018
- **Physics and Astronomy Colloquium**, *Sejong University*, "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", 11/21/2018
- **8th KIAS Workshop on Cosmology and Structure Formation**, *Korea Institute for Advanced Study*, "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", 11/06/2018
- **99th Korean Astronomical Society Meeting**, *hosted by KAS*, "Towards the Understanding of the Growth and Evolution of Supermassive Black Holes at Galaxy Centers", 10/10/2018

- **Physics Seminar**, *Korea Institute for Advanced Study*, “Dark Matter: A Computational Astrophysicist’s Perspective”, 10/08/2018
- **Particle Physics Korea Meeting**, *Seoul National University*, “Dark Matter: A Computational Astrophysicist’s Perspective”, 09/28/2018
- **Santa Cruz Galaxy Formation Workshop+7th AGORA Workshop**, *UC Santa Cruz*, “Overview of the AGORA High-resolution Cosmological Galaxy Simulations Comparison Project: 6 Years After Conception” (*joint with Joel Primack*), 08/10/2018
- **KDESci Meeting**, *Korea Institute for Advanced Study*, “Insights from the AGORA High-resolution Galaxy Simulations Comparison”, 07/09/2018
- **Galaxies A to Z Workshop**, *hosted by Center for Galaxy Evolution Research, Yonsei University*, “Studying the Interaction of Supermassive Black Holes and Their Host Galaxies”, 06/26/2018
- **Astronomy Colloquium**, *Seoul National University*, “Upcoming Era in Numerical Galaxy Formation: New Possibilities and Challenges”, 05/24/2018
- **Santa Cruz Galaxy Formation Workshop+6th AGORA Workshop**, *UC Santa Cruz*, “Reproducibility: An Insight from the AGORA High-resolution Galaxy Simulations Comparison”, 08/11/2017
- **Einstein Fellows Symposium 2016**, *Harvard-Smithsonian Center for Astrophysics*, “Reproducibility: An Insight from the AGORA High-resolution Galaxy Simulations Comparison”, 10/18/2016
- **Santa Cruz Galaxy Formation Workshop+5th AGORA Workshop**, *UC Santa Cruz*, “The AGORA High-resolution Galaxy Simulations Comparison. II: Isolated Disk Test - Kickoff Discussion”, 08/12/2016
- **Einstein Fellows Symposium 2015**, *Harvard-Smithsonian Center for Astrophysics*, “Upcoming New Era in Numerical Galaxy Formation: New Challenges and Possibilities”, 10/27/2015
- **Santa Cruz Galaxy Formation Workshop+4th AGORA Workshop**, *UC Santa Cruz*, “AGORA Initiative and Infrastructure: Where We Stand and Why We Are Here”, 08/21/2015
- **KIPAC Tea Talk**, *SLAC National Accelerator Laboratory*, “Challenges in Numerical Galaxy Formation and the AGORA Initiative”, 07/24/2015
- **Pasadena Astronomy Postdoc Symposium 2015**, *UCLA Lake Arrowhead Conference Center*, “Challenges in Numerical Galaxy Formation and the AGORA Initiative”, 04/09/2015
- **Astronomy Theory Postdoc Lunch**, *California Institute of Technology*, “AGORA High-resolution Galaxy Simulations Comparison Project”, 01/22/2015
- **Astronomy Colloquium**, *Seoul National University*, “Galaxy Formation Simulations in the High-resolution Era: Success and Challenge”, 12/10/2014
- **Einstein Fellows Symposium 2014**, *Harvard-Smithsonian Center for Astrophysics*, “Challenges in Numerical Galaxy Formation and the AGORA Initiative”, 10/28/2014
- **Santa Cruz Galaxy Formation Workshop+3rd AGORA Workshop**, *UC Santa Cruz*, “Status of the AGORA High-resolution Cosmological Galaxy Simulations Comparison Project: 2 Years After Conception” (*joint with Joel Primack*), 08/15/2014
- **Carnegie Observatories Colloquium Series**, *Carnegie Observatories*, “Rethinking Galaxy Simulations in the High-resolution Era”, 03/18/2014
- **Santa Cruz Galaxy Formation Workshop+2nd AGORA Workshop**, *UC Santa Cruz*, “The AGORA Project: Initial Conditions and the Proof-of-concept Test”, 08/16/2013
- **Cosmology Seminar**, *Stanford University*, “Rethinking Galaxy Simulations in the High-resolution Era”, 05/20/2013

- **Santa Cruz Galaxy Formation Workshop+Starting Workshop for the AGORA High-resolution Galaxy Simulations Comparison Project**, *UC Santa Cruz*, “Galaxy Formation with Radiating Molecular Cloud Particles”, 08/17/2012
- **Workshop for Korean Young Cosmologists**, *Korea Astronomy and Space Science Institute*, “High-resolution Galaxy Formation with Massive Black Holes and Radiating Star Clusters”, 06/25/2012
- **SLAC Association for Student Seminars**, *SLAC National Accelerator Laboratory*, 05/11/2011
- **KIPAC Tea Talk**, *SLAC National Accelerator Laboratory*, “Galaxy Formation and Mergers with Self-consistently Modeled Stars and Massive Black Holes”, 04/08/2011
- **Friday Lunchtime Astrophysics Seminar**, *UC Santa Cruz*, “Towards An Unabridged Understanding of The Coevolution of Galaxies and Massive Black Holes: What Have Simulators Tried? Why So Hard?”, 12/10/2010
- **Computational Astrophysics Group Seminar**, *University of Chicago*, “Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution”, 12/03/2010
- **Computational Cosmology and Galaxy Formation Seminar**, *Princeton University*, “Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution”, 11/15/2010
- **Theoretical Astrophysics Center Seminar**, *UC Berkeley*, “Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution”, 10/25/2010
- **Cosmoclub Seminar**, *UC Santa Cruz*, “Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution”, 10/18/2010
- **Friday Astro Lunch**, *UC Santa Barbara*, “Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution”, 10/15/2010
- **Astronomy Tea Talk**, *California Institute of Technology*, “Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes: Towards An Unabridged Understanding of Their Coevolution”, 10/11/2010
- **LCA Group Seminar**, *UC San Diego*, “Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes”, 09/30/2010
- **Santa Cruz Galaxy Formation Workshop**, *UC Santa Cruz*, “Galaxy Formation with Self-consistently Modeled Stars and Massive Black Holes”, 08/17/2010
- **HIPACC Summer School on Galaxy Formation**, *University of California High-Performance AstroComputing Center*, “Galaxy Formation using Enzo with Properly Modeled Stars and Massive Black Holes”, 07/28/2010